



earphones, or 32-ohm 40mm speakers, as shown in Fig. 14.

The volume level provided is

more than sufficient for listening through earphones or headphones.

For additional audio power requirements, you can connect an external power amplifier.

Once you have made all the necessary connections as per the circuit, insert the programmed ESP-01 into the circuit. Connect the earphones/headphones or the 32-ohm speaker to the audio output and attach the antenna wire to the FM module's pin 10.

Working prototype with MSOP-10 ZIF socket with

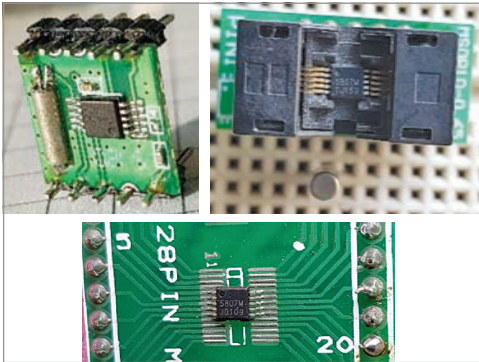


Fig. 13: FM IC mounted on module and 2mm header



Fig. 14: Earphones and speaker that can be used

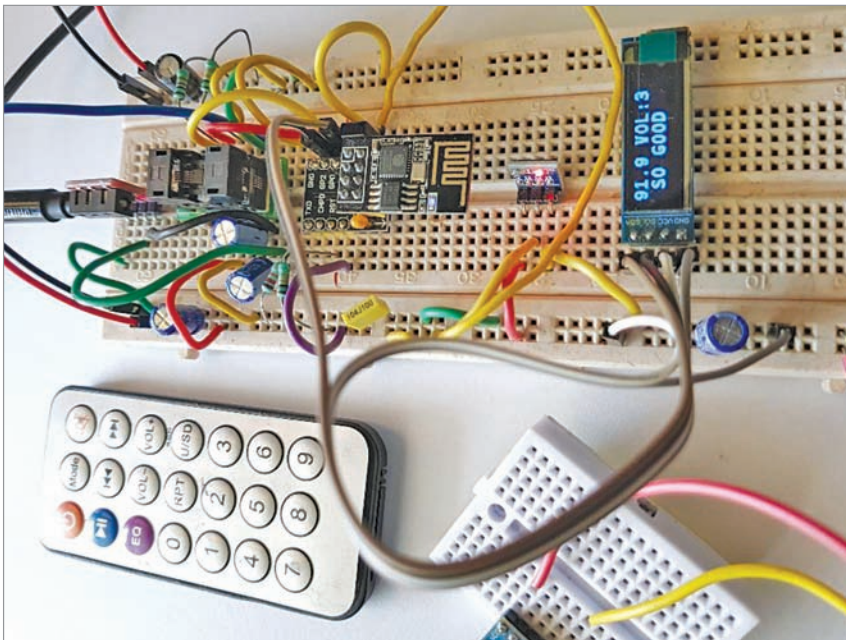


Fig. 15: Working proto with MSOP-10 ZIF socket with 2.54mm pitch along with IR receiver, ESP-01, and OLED with RDS display

2.54mm pitch along with IR receiver, ESP-01, and OLED with RDS display is shown in Fig. 15.

For testing, apply either a 5V or 3.7V power supply. You can change the volume using the remote control's VOL + and VOL- buttons and adjust the FM channel frequency using the < < (PREVIOUS) and > > (NEXT) buttons on the remote. Please refer to the NEC code table and the source code for key assignments.

Currently, the remote supports only four functions, but additional features like MUTE, BASS BOOST, and RSSI functions can be easily added, along with modifications to the display for status indications. You can also assign PRESET stations to remote keys 1, 2, 3, and so on, and store last-used station, volume information, and other status details in internal or external non-volatile memory. This can help boot the radio with the last-used volume and channel frequency information.

Moreover, if you have FM stereo transmitters, you can transmit and receive your own channel with stereo audio signals. This opens the possibility of operating FM remote speakers similar to Bluetooth speakers for wireless convenience.

**EFY Note.** Please note that not all FM stations transmit RDS information. The OLED display will indicate the RDS message if available from the station; otherwise, it will remain blank. **EFY**

The author K.N. Antony, with a B.Sc. degree in physics and a B.Tech degree in electronics, holds several US patents and is author of many international publications, including WIPO. His hobbies include RF and audio amplifier design

